

Characterization of incentive compatibility

- One may restrict attention to direct mechanisms in which truthful revelation is an equilibrium.
- Formalization: agents must have incentives to report truthfully.
- Model: IPVM, $N = 1$ object, I bidders.
- $X_i = [\underline{x}_i, \bar{x}_i]$ is i 's type space, $X = \prod_{i=1}^I X_i$ is the type-profile space. For easy of exposition we will use $X_i = [0, 1]$.
- A direct mechanism is $\{(p_i, t_i)\}_{i=1}^I$ where

$$p_i : X \rightarrow [0, 1], \quad (\forall x) \sum p_i(x) \leq 1$$
$$t_i : X \rightarrow \mathbb{R}$$

Ex post utility

- Fix a mechanism $\{p_i, t_i\}$.
- The player that observes x_i and reports x'_i when other agents observe x_{-i} obtains a payoff

$$u_i(x'_i | x_i, x_{-i}) = p_i(x'_i, x_{-i}) \cdot x_i - t_i(x'_i, x_{-i}).$$

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Interim utility

- Fix a mechanism $\{p_i, t_i\}_{i=1}^I$.
- If i observes $x_i \in X_i$, reports x'_i , but does not observe her opponents types x_{-i} , i 's expected payoff is

$$\underbrace{E_{x_{-i}} u_i(x'_i | x_i, x_{-i})}_{U_i(x'_i | x_i)} = \underbrace{E_{x_{-i}} p_i(x'_i, x_{-i})}_{Q_i(x'_i)} \cdot x_i - \underbrace{E_{x_{-i}} t_i(x'_i, x_{-i})}_{T_i(x'_i)}$$

$$U_i(x'_i | x_i) := Q_i(x'_i) \cdot x_i - T_i(x'_i)$$

$$U_i(x_i) := Q_i(x_i) \cdot x_i - T_i(x_i)$$

- The expectation is taken with respect to the *correct* probability distribution for x_{-i} . This will lead to a Bayesian Nash equilibrium.

Incentive compatibility (IC)

Definition 1 (Ex post IC) $\{p_i, t_i\}$ satisfies ex post incentive compatibility if $\forall (x_i, x_{-i}) \in X \forall x'_i \in X_i, u_i(x_i | x_i, x_{-i}) \geq u_i(x'_i | x_i, x_{-i})$.

Definition 2 (Interim IC) $\{p_i, t_i\}$ satisfies interim incentive compatibility if $\forall x_i \in X_i, x'_i \in X_i, U_i(x_i) \geq U_i(x'_i | x_i)$.

Note

- Ex post IC is often called dominant-strategy IC (DIC) or strategy-proofness.
- Interim IC is often called Bayesian IC (BIC); agent i chooses an action after observing her type but without observing other agents' types.
- There is also ex ante incentive compatibility.

See Myerson (1979), Myerson (1985).

Remarks

- $Q_i(x_i)$ is type x_i 's expected probability of trade.
- $T_i(x_i)$ is type x_i 's expected transfer to seller.
- DIC is determined by $\{(p_i, t_i)\}_{i=1}^I$.
- BIC is determined by $\{(Q_i, T_i)\}_{i=1}^I$.
 - Different mechanisms $\{(p_i, t_i)\}_{i=1}^I$ may yield the same (Q_i, T_i) .
Thus, they may all satisfy BIC but not DIC.
- Both BIC and DIC have the same mathematical structure.
 - In DIC x_{-i} is fixed and the defining inequality must hold for all x_{-i} .

Characterization

- Suppose there is a single agent, that is $I = 1$. Then
 - Subindex i need not be specified.
 - BIC and DSIC are equivalent.
- See Myerson (1981).
- See Rochet (1985), Rochet (1987) for an extension to N -good environments.

Theorem 1 IPV. Let $\{(p_i, t_i)\}_1^I$ be a direct mechanism and $\{(Q_i, T_i)\}_1^I$ be the respectively expected probability of trade (or expected allocation) and the expected transfer. Let $U_i(x_i) = Q_i(x_i) \cdot x_i - T_i(x_i)$.

The following statements are equivalent.

- a. $\{(p, t)\}_1^I$ satisfies Bayesian IC.
- b. $(\forall i)$ U_i is convex, $\frac{dU(x_i)}{dx_i} = Q_i(x_i) \in [0, 1]$ a.e.
- c. $(\forall i)$ $[Q_i(x'_i) - Q_i(x_i)] \cdot (x'_i - x_i) \geq 0$, and $Q_i(x_i) = \frac{dU(x_i)}{dx_i}$ a.e.
- d. $(\forall i)$ $\left[Q_i(x_i) \text{ is non-decreasing and } \right.$

$$\left. (\forall x_i) T_i(x_i) = Q_i(x_i)x_i - \int_{\underline{x}_i}^{x_i} Q_i(r) dr \right].$$

Corollary 1 Convexity of U_i implies that

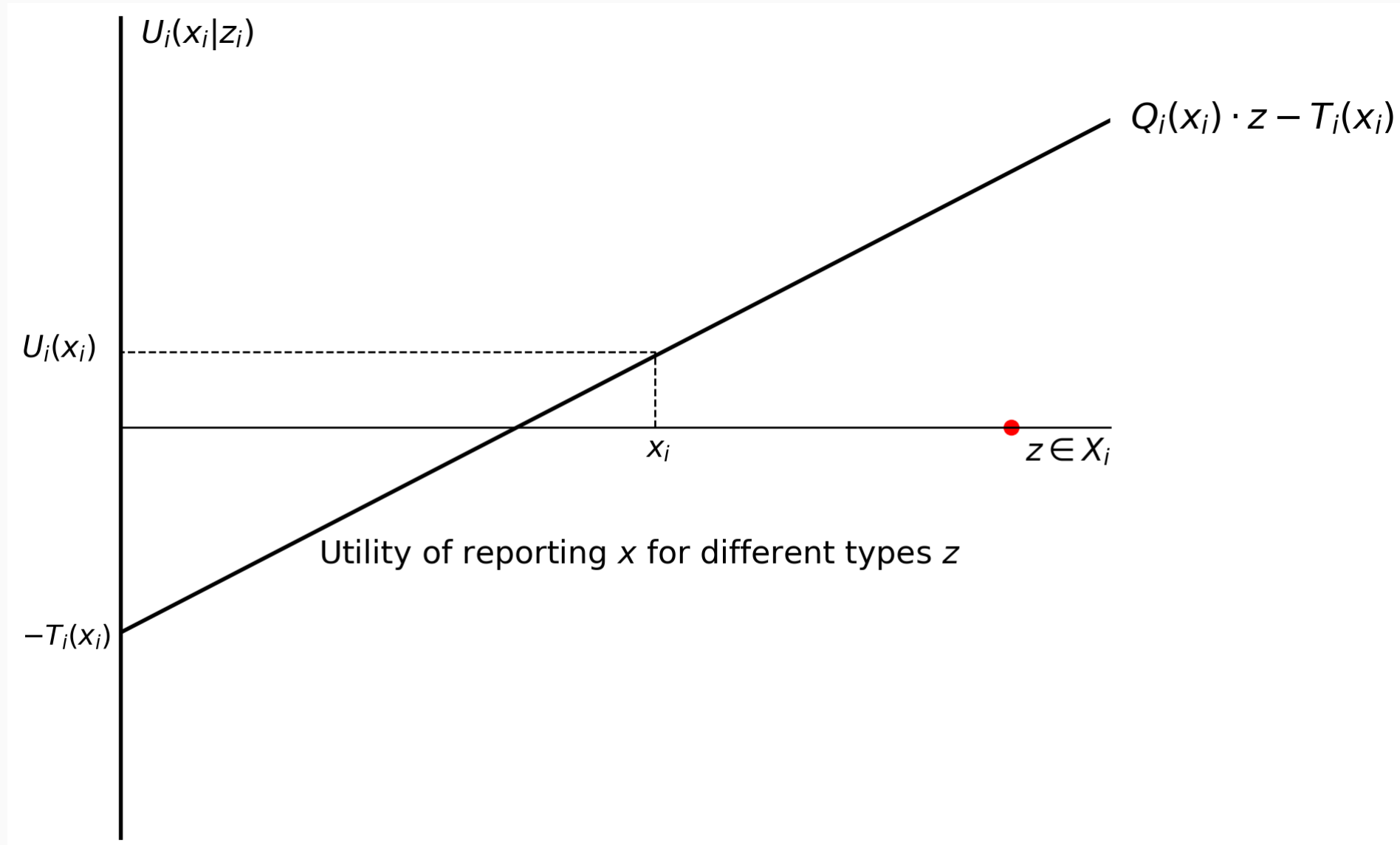
$$\begin{aligned} U_i(x_i) &= U_i(0) + \int_0^{x_i} \frac{dU_i(z)}{dz} dz \\ &= U_i(0) + \int_0^x Q_i(z) dz \end{aligned}$$

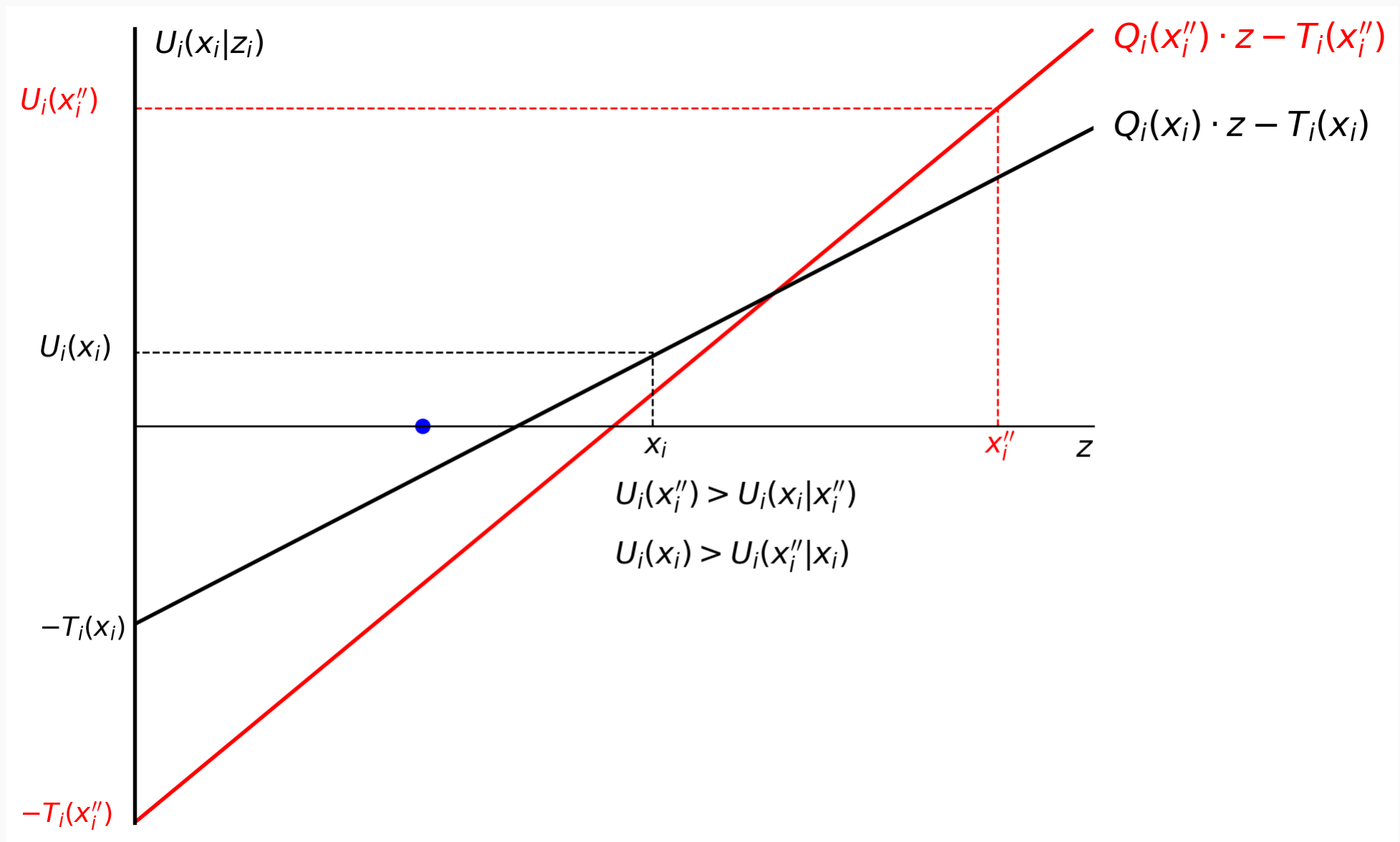
The Fundamental Theorem of Calculus holds for convex functions on the real line.

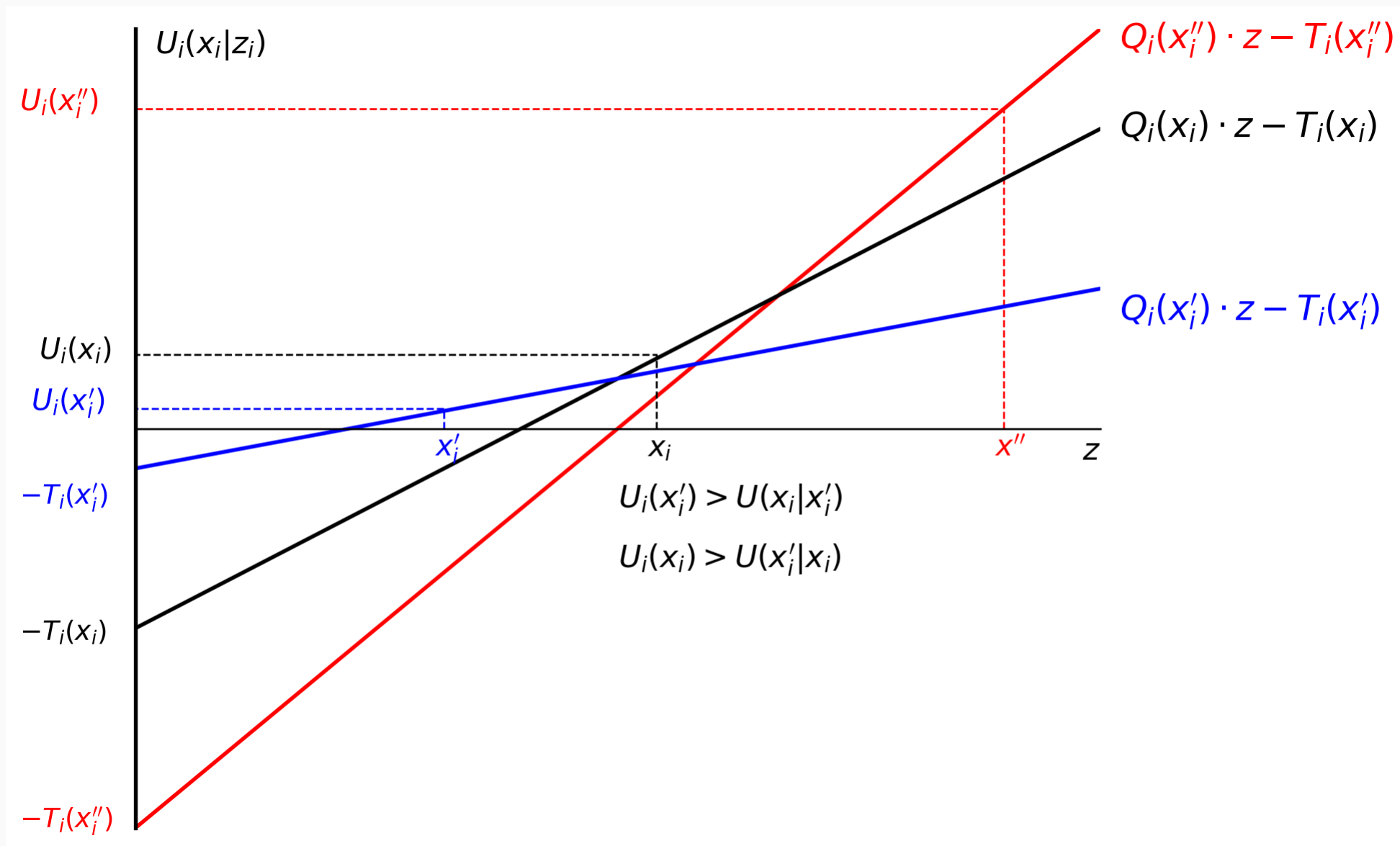
Because of convexity U_i is differentiable in all but a countable number of points. This implies that FTC holds.

→

$$a \implies b$$







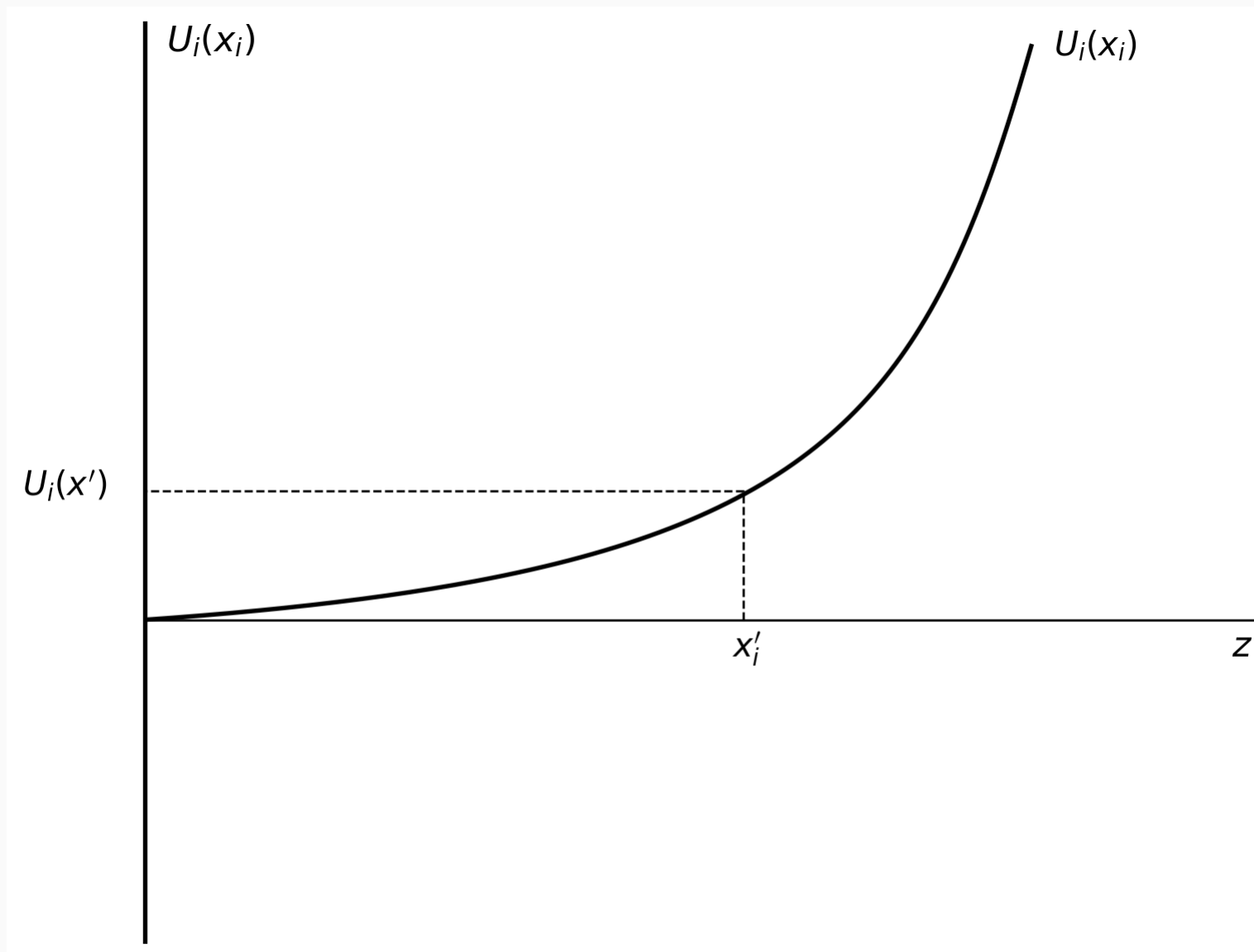
For all reports x'_i

$$\begin{aligned}U_i(x_i) &\geq U_i(x'_i|x_i) \\Q_i(x_i) \cdot x_i - T_i(x_i) &\geq Q_i(x'_i) \cdot x_i - T(x'_i) \\U_i(x_i) &= \sup_{x'_i} U_i(x'_i|x_i)\end{aligned}$$

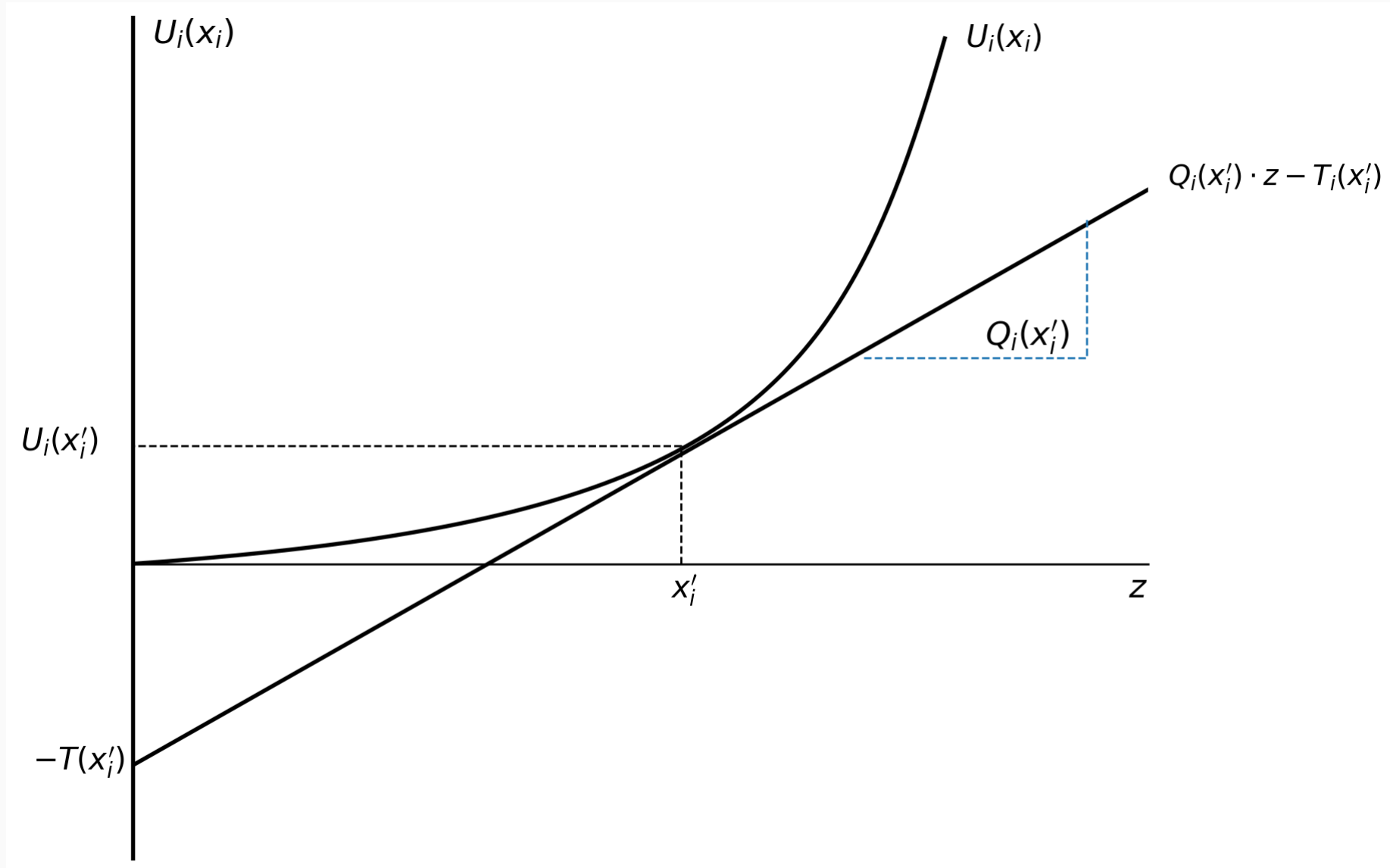
This implies u is convex because the pointwise supremum of a family of convex functions is convex. See for instance Rockafellar (1970).

Since convex functions defined on $[0, 1]$ are integrable, and $\frac{dU_i(x_i)}{dx_i} = Q_i(x_i)$ a.e.

Intuition reverse



Intuition reverse 2



$$b \implies c$$

$$U_i \text{ convex} \implies \begin{cases} U_i(x'_i) \geq U_i(x_i) + \frac{dU_i(x_i)}{dx_i} \cdot (x'_i - x_i) \\ U_i(x_i) \geq U_i(x'_i) - \frac{dU_i(x'_i)}{dx'_i} \cdot (x'_i - x_i) \end{cases}$$

- $U_i(x'_i) - U_i(x_i) \geq \underbrace{\frac{dU_i(x_i)}{dx_i}}_{Q_i(x_i)} \cdot (x'_i - x_i)$
- $\underbrace{\frac{dU_i(x'_i)}{dx'_i}}_{Q_i(x'_i)} \cdot (x'_i - x_i) \geq U_i(x'_i) - U_i(x_i).$
- then (c) $[Q_i(x'_i) - Q_i(x_i)] \cdot (x'_i - x_i) \geq 0 \quad a.e.$

$$c \implies a$$

- Property of convex functions,

$$U_i(x'_i) = U_i(x_i) + \int_{x_i}^{x'_i} Q_i(z) dz \quad (1)$$

Since $U_i(x_i) = Q_i(x_i) \cdot x_i - T_i(x_i)$,

then $-T_i(x) = -Q_i(x_i) \cdot x_i + U_i(x_i)$.

Hence $U_i(x_i|x'_i) = Q_i(x_i) \cdot x'_i - \overbrace{T(x_i)}$ or

$$U_i(x_i|x'_i) = Q_i(x_i) \cdot (x'_i - x_i) + U_i(x_i) \quad (2)$$

$$\begin{aligned}
U_i(x'_i) - U_i(x_i|x'_i) &= U_i(x_i) + \int_{x_i}^{x'_i} Q_i(z) dz - Q_i(x_i)(x'_i - x_i) \\
&\quad - U_i(x_i) \\
&= \int_{x_i}^{x'_i} Q_i(z) dz - Q_i(x_i)(x'_i - x_i) \\
&= \int_{x_i}^{x'_i} Q_i(z) dz - Q_i(x_i) \underbrace{\int_{x_i}^{x'_i} dz}_{x'_i - x_i} \\
&= \int_{x_i}^{x'_i} [Q_i(z) - Q_i(x_i)] dz \geq 0
\end{aligned}$$

where the first line replaces $U_i(x'_i)$ with [Equation 1](#) and $U_i(x_i|x'_i)$ with [Equation 2](#).

- Don't forget (d). QED

Questions left

- Can one use $\{Q_i, T_i\}_{i=1}^I$ as a primitive for a mechanism?
- Extend IC characterization to more general environments such as non-linear preferences, etc.
- Bayesian versus Dominant Strategy Incentive Compatibility
- Detour

References

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